



9065 - Characterizing environmental effects on the formation and evolution of water in star-forming regions: Orion as a case study

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	HOPS 12 NIRSpec	NIRSpec IFU Spectroscopy	(9) HOPS-12_NIRSpec
	2	HOPS 124 NIRSpec	NIRSpec IFU Spectroscopy	(2) HOPS-124_NIRSpec
	3	HOPS 358 NIRSpec	NIRSpec IFU Spectroscopy	(3) HOPS-358_NIRSpec
	4	HOPS 188 NIRSpec	NIRSpec IFU Spectroscopy	(4) HOPS-188_NIRSpec
	5	HOPS 310 NIRSpec	NIRSpec IFU Spectroscopy	(13) HOPS-310_NIRSpec
	6	HOPS 321 NIRSpec	NIRSpec IFU Spectroscopy	(6) HOPS-321_NIRSpec
	7	HOPS 12 MIRI	MIRI Medium Resolution Spectroscopy	(1) HOPS-12_MIRI
	8	HOPS 124 MIRI	MIRI Medium Resolution Spectroscopy	(10) HOPS-124_MIRI
	9	HOPS 358 MIRI	MIRI Medium Resolution Spectroscopy	(11) HOPS-358_MIRI

Folder	Observation	Label	Observing Template	Science Target
	10	HOPS 188 MIRI	MIRI Medium Resolution Spectroscopy	(12) HOPS-188_MIRI
	11	HOPS 310 MIRI	MIRI Medium Resolution Spectroscopy	(5) HOPS-310_MIRI
	12	HOPS 321 MIRI	MIRI Medium Resolution Spectroscopy	(14) HOPS-321_MIRI
	13	MIRI N BKGD	MIRI Medium Resolution Spectroscopy	(7) Orion-bkgd-N
	14	MIRI S BKGD	MIRI Medium Resolution Spectroscopy	(8) Orion-bkgd-S

ABSTRACT

Water is essential to many of the physical processes that drive star and planet formation as well as to the existence of life on Earth. However, how local environmental conditions affect the formation and evolution of water in young star-forming regions remains poorly understood. Deducing these effects is vital to characterizing water's chemical evolution in protostellar objects and, subsequently, how it is accreted from protoplanetary disks by icy outer solar system bodies like comets and habitable rocky worlds like Earth.

To that end, we propose to measure the HDO/H₂O ratios in the ice envelopes of 6 isolated young protostars in the Orion molecular cloud complex, suggested to be analogous to the star-forming region where our solar system formed. The source sample was selected to complement other JWST programs measuring ice HDO/H₂O ratios in clustered Orion protostars, ALMA programs measuring HDO/H₂O in warm protostellar gas, and cometary HDO/H₂O measurements. In synergy with these works, the proposed observations will elucidate how local environmental parameters like proximity to other stellar objects affect the formation and chemical processing of water during the pre- and protostellar stages and, ultimately, clarify the veracity of the tentative link between the water in our solar system and the water ice from these most primitive epochs of star formation.

OBSERVING DESCRIPTION

We propose to quantify the HDO/H₂O ice ratio toward 6 isolated protostars in the Orion molecular cloud complex. The HDO ice column density will be obtained using the NIRSpec IFU G395M grating, which covers the HDO ice feature at 4.1 μ m. H₂O ice will be quantified using a combination of NIRSpec IFU G235M/PRISM and G395M spectra if the broad H₂O 3 μ m feature is not saturated, with full MIRI MRS spectra providing an alternative means of H₂O ice quantification via the 12 μ m H₂O libration band if the band is saturated, as is oftentimes the case toward embedded protostars. The proposed observations all utilize the recommended 4-dither patterns and require NIRSpec leakcal for 4 of the sources due to neighboring bright sources.

By comparing the HDO/H₂O ice ratios obtained from this program with those obtained from other JWST NIRSpec G395M data of clustered Orion sources as well as gas HDO/H₂O measurements of both clustered and isolated Orion sources from ALMA, the proposed observations will illuminate

how water ice deuterium ratios are affected by clustered vs. isolated environments and whether such differences in environments affect the degree to which water is processed once sublimated into the gas phase in protostellar envelopes.

Proposal 9065 - Targets - Characterizing environmental effects on the formation and evolution of water in star-forming regions: Orion ...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	HOPS-12_MIRI	RA: 05 35 8.5968 (83.7858200d) Dec: -05 55 53.86 (-5.93163d) Equinox: J2000	Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES				
	(2)	HOPS-124_NIRSpec	RA: 05 39 19.9800 (84.8332500d) Dec: -07 26 11.20 (-7.43644d) Equinox: J2000	Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES				
	(3)	HOPS-358_NIRSpec	RA: 05 46 7.2600 (86.5302500d) Dec: -00 13 30.23 (-.22506d) Equinox: J2000	Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES				
	(4)	HOPS-188_NIRSpec	RA: 05 35 29.8503 (83.8743762d) Dec: -06 26 58.33 (-6.44954d) Equinox: J2000	Proper Motion RA: -0.861 mas/yr Proper Motion Dec: 2.306 mas/yr Parallax: 0.0037729" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES				
	(5)	HOPS-310_MIRI	RA: 05 42 27.6800 (85.6153333d) Dec: -01 20 1.00 (-1.33361d) Equinox: J2000	Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES				
	(6)	HOPS-321_NIRSpec	RA: 05 46 33.1880 (86.6382833d) Dec: +00 00 2.15 (.00060d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES					

Proposal 9065 - Targets - Characterizing environmental effects on the formation and evolution of water in star-forming regions: Orion ...

(7)	Orion-bkgd-N	RA: 05 46 32.7000 (86.6362500d) Dec: +00 14 39.00 (.24417d) Equinox: J2000	
Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES			
(8)	Orion-bkgd-S	RA: 05 36 56.2500 (84.2343750d) Dec: -07 10 12.03 (-7.17001d) Equinox: J2000	
Comments: Category=ISM Description=[Dense interstellar clouds, Intergalactic medium, Interstellar clouds, Interstellar dust] Extended=YES			
(9)	HOPS-12_NIRSpec	RA: 05 35 8.5968 (83.7858200d) Dec: -05 55 53.86 (-5.93163d) Equinox: J2000	Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES			
(10)	HOPS-124_MIRI	RA: 05 39 19.9800 (84.8332500d) Dec: -07 26 11.20 (-7.43644d) Equinox: J2000	Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES			
(11)	HOPS-358_MIRI	RA: 05 46 7.2600 (86.5302500d) Dec: -00 13 30.23 (-.22506d) Equinox: J2000	Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES			
(12)	HOPS-188_MIRI	RA: 05 35 29.8503 (83.8743762d) Dec: -06 26 58.33 (-6.44954d) Equinox: J2000	Proper Motion RA: -0.861 mas/yr Proper Motion Dec: 2.306 mas/yr Parallax: 0.0037729" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES			
(13)	HOPS-310_NIRSpec	RA: 05 42 27.6800 (85.6153333d) Dec: -01 20 1.00 (-1.33361d) Equinox: J2000	Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES			

Proposal 9065 - Targets - Characterizing environmental effects on the formation and evolution of water in star-forming regions: Orion ...

(14)	HOPS-321_MIRI	RA: 05 46 33.1880 (86.6382833d) Dec: +00 00 2.15 (.00060d) Equinox: J2000	Epoch of Position: 2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=ISM</i> <i>Description=[Interstellar clouds, Interstellar dust, Protostars]</i> <i>Extended=YES</i>
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Proposal 9065 - Observation 1 - Characterizing environmental effects on the formation and evolution of water in star-forming regions: ...

Observation	Proposal 9065, Observation 1: HOPS 12 NIRSpec											Tue Mar 11 23:06:58 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(9)	HOPS-12_NIRSpec	RA: 05 35 8.5968 (83.7858200d) Dec: -05 55 53.86 (-5.93163d) Equinox: J2000				Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395M/F290LP	NRSIRS2RAPID	45	2	false	true	NONE	4	8	5368.712	218001
	2	G395M/F290LP	NRSIRS2RAPID	45	2	true	false	NONE	1	2	1342.178	218001

Proposal 9065 - Observation 2 - Characterizing environmental effects on the formation and evolution of water in star-forming regions: ...

Observation	Proposal 9065, Observation 2: HOPS 124 NIRSpec											Tue Mar 11 23:06:58 GMT 2025
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	HOPS-124_NIRSpec	RA: 05 39 19.9800 (84.8332500d) Dec: -07 26 11.20 (-7.43644d) Equinox: J2000				Epoch of Position: 2000					
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES												
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	NRSIRS2RAPID	10	1	false	true	NONE	4	4	641.911	218001
	2	G395M/F290LP	NRSIRS2RAPID	10	1	false	true	NONE	4	4	641.911	218001

Proposal 9065 - Observation 3 - Characterizing environmental effects on the formation and evolution of water in star-forming regions: ...

Observation	Proposal 9065, Observation 3: HOPS 358 NIRSpec												Tue Mar 11 23:06:58 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	HOPS-358_NIRSpec		RA: 05 46 7.2600 (86.5302500d)			Epoch of Position: 2000						
				Dec: -00 13 30.23 (-.22506d)									
				Equinox: J2000									
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES											
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	4	4	350.133	218001	
	2	PRISM/CLEAR	NRSIRS2RAPID	5	1	true	false	NONE	1	1	87.533	218001	
	3	G395M/F290LP	NRSIRS2RAPID	15	1	false	true	NONE	4	4	933.689	218001	
	4	G395M/F290LP	NRSIRS2RAPID	15	1	true	false	NONE	1	1	233.422	218001	

Proposal 9065 - Observation 4 - Characterizing environmental effects on the formation and evolution of water in star-forming regions: ...

Observation	Proposal 9065, Observation 4: HOPS 188 NIRSpec											Tue Mar 11 23:06:58 GMT 2025
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	HOPS-188_NIRSpec	RA: 05 35 29.8503 (83.8743762d) Dec: -06 26 58.33 (-6.44954d) Equinox: J2000				Proper Motion RA: -0.861 mas/yr Proper Motion Dec: 2.306 mas/yr Parallax: 0.0037729" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	NRSIRS2RAPID	5	1	false	true	NONE	4	4	350.133	218001
	2	G395M/F290LP	NRSRAPID	4	2	false	true	NONE	4	8	429.471	218001

Proposal 9065 - Observation 5 - Characterizing environmental effects on the formation and evolution of water in star-forming regions: ...

Observation	Proposal 9065, Observation 5: HOPS 310 NIRSpec											Tue Mar 11 23:06:58 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(13)	HOPS-310_NIRSpec	RA: 05 42 27.6800 (85.6153333d)				Epoch of Position: 2000					
			Dec: -01 20 1.00 (-1.33361d)									
			Equinox: J2000									
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	4	4	350.133	218001
	2	PRISM/CLEAR	NRSIRS2RAPID	5	1	true	false	NONE	1	1	87.533	218001
	3	G395M/F290LP	NRSIRS2RAPID	50	1	false	true	NONE	4	4	2976.134	218001
	4	G395M/F290LP	NRSIRS2RAPID	50	1	true	false	NONE	1	1	744.033	218001

Proposal 9065 - Observation 6 - Characterizing environmental effects on the formation and evolution of water in star-forming regions: ...

Observation	Proposal 9065, Observation 6: HOPS 321 NIRSpec												Tue Mar 11 23:06:58 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(6)	HOPS-321_NIRSpec		RA: 05 46 33.1880 (86.6382833d)			Epoch of Position: 2000						
				Dec: +00 00 2.15 (.00060d)									
				Equinox: J2000									
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
Template	Category=ISM												
	Description=[Interstellar clouds, Interstellar dust, Protostars]												
	Extended=YES												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	G235M/F170LP	NRSIRS2RAPID	10	1	false	true	NONE	4	4	641.911	218001	
	2	G235M/F170LP	NRSIRS2RAPID	10	1	true	false	NONE	1	1	160.478	218001	
	3	G395M/F290LP	NRSIRS2RAPID	10	1	false	true	NONE	4	4	641.911	218001	
	4	G395M/F290LP	NRSIRS2RAPID	10	1	true	false	NONE	1	1	160.478	218001	

Proposal 9065 - Observation 7 - Characterizing environmental effects on the formation and evolution of water in star-forming regions: ...

Observation	Proposal 9065, Observation 7: HOPS 12 MIRI												Tue Mar 11 23:06:58 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI S BKGD (Obs 14)]												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	HOPS-12_MIRI	RA: 05 35 8.5968 (83.7858200d) Dec: -05 55 53.86 (-5.93163d) Equinox: J2000				Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=ISM												
	Description=[Interstellar clouds, Interstellar dust, Protostars]												
Acquisition	Extended=YES												
	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				NO			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SHORT(A)	MRSLONG		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518
	1	SHORT(A)	MRSSHORT		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518
	2	MEDIUM(B)	MRSLONG		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518
	2	MEDIUM(B)	MRSSHORT		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518
	3	LONG(C)	MRSLONG		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518
	3	LONG(C)	MRSSHORT		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518

Special Requirements	Sequence Observations 7, 8, 10, 14, Non-interruptible
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Proposal 9065 - Observation 8 - Characterizing environmental effects on the formation and evolution of water in star-forming regions: ...

Observation	Proposal 9065, Observation 8: HOPS 124 MIRI												Tue Mar 11 23:06:59 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI S BKGD (Obs 14)]													
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(10)	HOPS-124_MIRI			RA: 05 39 19.9800 (84.8332500d) Dec: -07 26 11.20 (-7.43644d) Equinox: J2000			Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=ISM													
	Description=[Interstellar clouds, Interstellar dust, Protostars]													
Extended=YES														
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			NO			FULL			Allow Auto Reorder			
Dithers	#	Dither Type			Optimized For			Direction						
	1	4-Point			EXTENDED SOURCE			NEGATIVE						
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	90	1	1	Dither 1	4	4	999.014	225518	
	1	SHORT(A)	MRSSHORT		FASTR1	90	1	1	Dither 1	4	4	999.014	225518	
	2	MEDIUM(B)	MRSLONG		FASTR1	90	1	1	Dither 1	4	4	999.014	225518	
	2	MEDIUM(B)	MRSSHORT		FASTR1	90	1	1	Dither 1	4	4	999.014	225518	
	3	LONG(C)	MRSLONG		FASTR1	90	1	1	Dither 1	4	4	999.014	225518	
	3	LONG(C)	MRSSHORT		FASTR1	90	1	1	Dither 1	4	4	999.014	225518	

Special Requirements	Sequence Observations 7, 8, 10, 14, Non-interruptible
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Proposal 9065 - Observation 9 - Characterizing environmental effects on the formation and evolution of water in star-forming regions: ...

Observation	Proposal 9065, Observation 9: HOPS 358 MIRI												Tue Mar 11 23:06:59 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI N BKGD (Obs 13)]												
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(11)	HOPS-358_MIRI		RA: 05 46 7.2600 (86.5302500d)			Epoch of Position: 2000						
				Dec: -00 13 30.23 (-.22506d)									
				Equinox: J2000									
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES											
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS			NO			FULL			Allow Auto Reorder		
Dithers	#	Dither Type			Optimized For			Direction					
	1	4-Point			EXTENDED SOURCE			NEGATIVE					
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SHORT(A)	MRSLONG		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518
	1	SHORT(A)	MRSSHORT		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518
	2	MEDIUM(B)	MRSLONG		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518
	2	MEDIUM(B)	MRSSHORT		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518
	3	LONG(C)	MRSLONG		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518
	3	LONG(C)	MRSSHORT		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518

Special Requirements	Sequence Observations 9, 11, 12, 13, Non-interruptible
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Proposal 9065 - Observation 10 - Characterizing environmental effects on the formation and evolution of water in star-forming regions:...

Observation	Proposal 9065, Observation 10: HOPS 188 MIRI												Tue Mar 11 23:06:59 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI S BKGD (Obs 14)]												
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(12)	HOPS-188_MIRI	RA: 05 35 29.8503 (83.8743762d) Dec: -06 26 58.33 (-6.44954d) Equinox: J2000				Proper Motion RA: -0.861 mas/yr Proper Motion Dec: 2.306 mas/yr Parallax: 0.0037729" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=ISM												
	Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				NO			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wbk.Calc ID
	1	SHORT(A)	MRSLONG		FASTR1	35	1	1	Dither 1	4	4	388.506	225518
	1	SHORT(A)	MRSSHORT		FASTR1	35	1	1	Dither 1	4	4	388.506	225518
	2	MEDIUM(B)	MRSLONG		FASTR1	35	1	1	Dither 1	4	4	388.506	225518
	2	MEDIUM(B)	MRSSHORT		FASTR1	35	1	1	Dither 1	4	4	388.506	225518
	3	LONG(C)	MRSLONG		FASTR1	35	1	1	Dither 1	4	4	388.506	225518
	3	LONG(C)	MRSSHORT		FASTR1	35	1	1	Dither 1	4	4	388.506	225518
	3	LONG(C)	MRSSHORT		FASTR1	35	1	1	Dither 1	4	4	388.506	225518

Special Requirements	Sequence Observations 7, 8, 10, 14, Non-interruptible
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Proposal 9065 - Observation 11 - Characterizing environmental effects on the formation and evolution of water in star-forming regions:...

Observation	Proposal 9065, Observation 11: HOPS 310 MIRI													Tue Mar 11 23:06:59 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI N BKGD (Obs 13)]													
Diagnosics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(5)	HOPS-310_MIRI	RA: 05 42 27.6800 (85.6153333d) Dec: -01 20 1.00 (-1.33361d) Equinox: J2000				Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=ISM													
	Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				NO				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	35	1	1	Dither 1	4	4	388.506	225518	
	1	SHORT(A)	MRSSHORT		FASTR1	35	1	1	Dither 1	4	4	388.506	225518	
	2	MEDIUM(B)	MRSLONG		FASTR1	35	1	1	Dither 1	4	4	388.506	225518	
	2	MEDIUM(B)	MRSSHORT		FASTR1	35	1	1	Dither 1	4	4	388.506	225518	
	3	LONG(C)	MRSLONG		FASTR1	35	1	1	Dither 1	4	4	388.506	225518	
	3	LONG(C)	MRSSHORT		FASTR1	35	1	1	Dither 1	4	4	388.506	225518	

Special Requirements	Sequence Observations 9, 11, 12, 13, Non-interruptible
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Proposal 9065 - Observation 12 - Characterizing environmental effects on the formation and evolution of water in star-forming regions:...

Observation	Proposal 9065, Observation 12: HOPS 321 MIRI													Tue Mar 11 23:06:59 GMT 2025	
	Diagnostic Status: Warning														
	Observing Template: MIRI Medium Resolution Spectroscopy														
	Background Observations:[MIRI N BKGD (Obs 13)]														
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.														
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous				
	(14)	HOPS-321_MIRI			RA: 05 46 33.1880 (86.6382833d)			Epoch of Position: 2000							
					Dec: +00 00 2.15 (.00060d)										
					Equinox: J2000										
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar clouds, Interstellar dust, Protostars] Extended=YES													
Acquisition	#	Target													
	1	NONE													
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
		All MRS			NO			FULL			Allow Auto Reorder				
Dithers	#	Dither Type			Optimized For			Direction							
	1	4-Point			EXTENDED SOURCE			NEGATIVE							
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID		
	1	SHORT(A)	MRSLONG		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518		
	1	SHORT(A)	MRSSHORT		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518		
	2	MEDIUM(B)	MRSLONG		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518		
	2	MEDIUM(B)	MRSSHORT		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518		
	3	LONG(C)	MRSLONG		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518		
	3	LONG(C)	MRSSHORT		FASTR1	90	2	1	Dither 1	4	8	2009.129	225518		

Special Requirements	Sequence Observations 9, 11, 12, 13, Non-interruptible
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Proposal 9065 - Observation 13 - Characterizing environmental effects on the formation and evolution of water in star-forming regions:...

Observation	Proposal 9065, Observation 13: MIRI N BKGD												Tue Mar 11 23:06:59 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [HOPS 358 MIRI (Obs 9), HOPS 310 MIRI (Obs 11), HOPS 321 MIRI (Obs 12)]													
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous				
	(7)	Orion-bkgd-N		RA: 05 46 32.7000 (86.6362500d) Dec: +00 14 39.00 (.24417d) Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			NO			FULL			Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	2-Point				BACKGROUND				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	90	2	1	Dither 1	2	4	1004.564		
	1	SHORT(A)	MRSSHORT		FASTR1	90	2	1	Dither 1	2	4	1004.564		
	2	MEDIUM(B)	MRSLONG		FASTR1	90	2	1	Dither 1	2	4	1004.564		
	2	MEDIUM(B)	MRSSHORT		FASTR1	90	2	1	Dither 1	2	4	1004.564		
	3	LONG(C)	MRSLONG		FASTR1	90	2	1	Dither 1	2	4	1004.564		
	3	LONG(C)	MRSSHORT		FASTR1	90	2	1	Dither 1	2	4	1004.564		

Special Requirements	Sequence Observations 9, 11, 12, 13, Non-interruptible
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Proposal 9065 - Observation 14 - Characterizing environmental effects on the formation and evolution of water in star-forming regions:...

Observation	Proposal 9065, Observation 14: MIRI S BKGD													Tue Mar 11 23:06:59 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [HOPS 12 MIRI (Obs 7), HOPS 124 MIRI (Obs 8), HOPS 188 MIRI (Obs 10)]													
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(8)	Orion-bkgd-S	RA: 05 36 56.2500 (84.2343750d) Dec: -07 10 12.03 (-7.17001d) Equinox: J2000											
	Comments: Category=ISM Description=[Dense interstellar clouds, Interqalactic medium, Interstellar clouds, Interstellar dust]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
		All MRS				NO			FULL		Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	2-Point					BACKGROUND			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	90	2	1	Dither 1	2	4	1004.564		
	1	SHORT(A)	MRSSHORT		FASTR1	90	2	1	Dither 1	2	4	1004.564		
	2	MEDIUM(B)	MRSLONG		FASTR1	90	2	1	Dither 1	2	4	1004.564		
	2	MEDIUM(B)	MRSSHORT		FASTR1	90	2	1	Dither 1	2	4	1004.564		
	3	LONG(C)	MRSLONG		FASTR1	90	2	1	Dither 1	2	4	1004.564		
	3	LONG(C)	MRSSHORT		FASTR1	90	2	1	Dither 1	2	4	1004.564		

Proposal 9065 - Observation 14 - Characterizing environmental effects on the formation and evolution of water in star-forming regions:...

Special Requirements	Sequence Observations 7, 8, 10, 14, Non-interruptible
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